



Mohamed Elkeiy

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Development engineer specializing in power electronics, hardware development, and control systems for HV converters and Electric Motors. Strong expertise in HV converter model-based development with international work experience across Germany, United States and Egypt.

Highly motivated to drive innovation in electromobility and power grid electronics applications..

Date/Place of birth

- 01.07.1996
- Alexandria, Egypt

Nationality

- Egyptian - German

Development Tools

Language Tools

- C, C++
- MATLAB scripting

Development Softwares

- MATLAB Simulink
- PSOC Creator
- LTSpice, Proteus
- Vector Tools
- PLC SIMATIC Manager

Micro-controllers

- DSP-TI C2000 family / AM265
- SR5-EI ST micro controllers
- PSOC-5LP
- ATMEL AVR

Industrial Controllers

- SIEMENS PLC S-300 and S-400
- SIEMENS Micromaster VFDs

CAD Tools

- Altium Designer, Mentor Graphics
- AutoCAD, CATIA V5

Certificates

- Functional safety in Automotive (FUSA) Certificate
- Classic Control Diploma
- MATLAB Certification from Mathworks
- SIEMENS course in Drives systems and power converters
- SIEMENS PLC S-300 Diploma

Languages

- Arabic (Native)
- English – C1 (Full Professional Proficiency)
- German – C1 (Full Professional Proficiency)

Experience

Senior Power Electronics Control Engineer of OBC Converters

Flex – Filderstadt, Germany | August 2025 – Present

- Responsible for Control design and SW implementation of high power FORD OBC Converter
- Model-based Implementation of Control Architecture for PFC-HVDCDC Converter stages
- V-Model development of OBC project – HIL rapid prototyping and testing environment for newly designed control concepts

Techs- ST Electronics Micro-controllers - MATLAB/SIMULINK – Vector tools (CANoe – V-Flash, CANape – CANdela) - LTSpice- Mentor Graphics - WaveStudio - Pig Picture Softwares (JIRA-JAMA-Confluence)-SAP- Lab equipments for HVDC systems (HV power racks, high voltage isolated equipment, CAN systems), HV voltage safety, first aid training

Firmware Power Electronics Engineer

MKS Inc – Wilmington, MA, USA | April 2025 – June 2025

- HVAC converter control for Plasma Source generators (RPS) used in Semiconductor Fablabs
- Optimization of control architect to align with increased power density requirements of RPS (Remote Plasma Source).

Techs- TI C2000 micro-controllers – TI tools (Code Composer, SFRA, Uniflash), LTSpice- MATLAB/SIMULINK - Lab equipments for HVDC systems (HV power racks, high voltage isolated equipment), HV voltage safety, first aid training

Power Electronics HW Development Manager of HVDCDC Converters

Panasonic Industrial Devices – Lüneburg, Germany | October 2023 – March 2025

- Power electronics simulation and design for high-voltage DCDC converters of Mercedes Benz EV cars.
- Responsible for electronics components supplier production change technical evaluations
- V-Model development of DCDC project – Hardware project manager HPM
- Planning Hardware team activities and tasks
- Responsible for Baseline of new changes in EV converter
- Responsible for production specifications of HVDCDC converters in Panasonic production sites

Techs- Mentor Graphics - CANoe – V-Flash, LTSpice- MATLAB/SIMULINK - MAUI - Pig Picture Softwares (JIRA-JAMA-Confluence)-SAP- Lab equipments for HVDC systems (HV power racks, high voltage isolated equipment, CAN systems), HV voltage safety, first aid training

Development Engineer for Power Tools

HILTI R&D Center – Kaufering, Bavaria, Germany | July 2022 – September 2023

- Power electronics hardware testing and Initial operation of power tools.
 - Circuit Simulation and validation of inverter gate drives.
 - Motor control development using sensorless FOC for Brushless DC Motor.
- Techs- HILTI tools - Pico Loggers- LTSpice- Altium Designer- STM32 ST-Link Utility - Keil - Visual Studio –Lab equipments (Oscilloscope, logic analyzer, multimeters, data loggers, temperature loggers)

College Projects

Remotely Operated Vehicles (ROV) competition

Description- Member of college technical team. Responsible for motor drives control and PCB design.

Model-based design of Electric Motor Drive System

Description- Vector Control of Induction Motor in Simulink environment using different control algorithms (FOC - MPC) and implementation with DSP micro controller

Research assistant at Faculty of Engineering

Research Project in Electric Vehicles Charging Topologies - Multi-port Charging System for EVs

Alexandria University – Alexandria, Egypt | January 2021 – June 2022

Description- Model Based Design of a charging converter. Mathematical modelling and simulation of the system (MIL phase).

- Design and implementation of the control using DSP and PSoC controllers.
- Model validation by building a lab set-up (HIL).
- DAB control using different control schemes (PI – MPC)

Techs- MATLAB Simulink, C, PSOC creator, Code composer, DSPACE Labbox, CANALYZER, Power electronics, double-pulse testing, PCB design and Microcontrollers, Professional lab equipments (Oscilloscope, logic analyzer, multimeters, LC meters, Soldering of SMD and DIP custom components)

Electrical Engineer Internship

BMW Motorcycle Safety, Innovation and riding assistance

BMW MOTORRAD - Munich, Bavaria, Germany | August 2020 – December 2020

Description- Measuring and calibration of strain gauge sensors that were used to measuring the steering torque in motorcycles

- Control of Motor actuators being used in Ergonomics Project
- Built a Simulink model for the CAN data output from the sensors and worked with DBC files using CANdb++

Techs- MATLAB Simulink, CANdb++, CAN loggers, CATIA

Education

MSc. in Electrical Engineering - Technical University of Munich (TUM), Germany | October 2019 – October 2022

- Major: Power Electronics, Control Systems, Electrical Engineering and Energy Systems
- Master Thesis: Design of MBD Topology for fast chargers of Electric Vehicles
- Grade: Distinction with honors (German GPA 1.8)

BSc. in Electrical Engineering – Alexandria University, Egypt | August 2014 – July 2019

- Major: Electronics, Motor Control and Drive Systems and Electrical Engineering
- Bachelor Thesis: Design of Three-phase Inverter for Induction Motors
- Grade: Distinction with honors 88.7%

Academic Publications

"Model Predictive Control for Dual Active Bridge in fast Charging Application of Electric Vehicles," - Main author (IEEE Journal)

"Multiport DC-DC Converter with Differential Power Processing for fast EV Charging Stations" - Main author (Sustainability Journal MDPI)